

Setup Sysmon

```
PS Y:\system_tools> .\Sysmon64.exe -i

System Monitor v15.15 - System activity monitor
By Mark Russinovich and Thomas Garnier
Copyright (C) 2014-2024 Microsoft Corporation
Using libxml2. libxml2 is Copyright (C) 1998-2012 Daniel Veillard. All Rights Reserved.
Sysinternals - www.sysinternals.com

The service Sysmon64 is already registered. Uninstall Sysmon before reinstalling.

PS Y:\system_tools>
```

```
PS Y:\system_tools> .\Sysmon64.exe -c .\Sysmon_Configuration.xml

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Loading configuration file with schema version 15.15
Sysmon schema version: 4.90
Configuration file validated.
Configuration updated.
```

Test cases

Run ps script to detect events

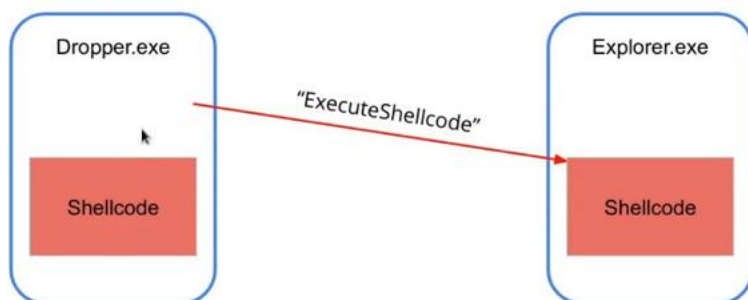
```
$time = Get-Date -Format "MM-dd-yyyy hh:mm:ss tt"
while($true)
{
    try{
        Get-WinEvent -FilterHashtable @{logname='Microsoft-Windows-Sysmon/Operational';StartTime=$time;id=8,10} -ErrorAction SilentlyContinue | ForEach-Object{
            write-host "[$($_.TimeCreated) - EventID: $($_.Id)]" -ForegroundColor Green
            write-host "$($_.Message)"
            write-host "-----"
        }
    }
    catch{
        Write-Host "Error caught: Nothing to display" -ForegroundColor Gray
    }
    $time = Get-Date
    Start-Sleep 3
}
```

Case 1 - Detect shell injection

Detection rules

```
<!--Event ID 8 CreateRemoteThread -->
<CreateRemoteThread onmatch="exclude">
  <SourceImage condition="is">C:\Windows\system32\svchost.exe</SourceImage>
  <SourceImage condition="is">C:\Windows\system32\wininit.exe</SourceImage>
  <SourceImage condition="is">C:\Windows\system32\csrss.exe</SourceImage>
  <SourceImage condition="is">C:\Windows\system32\services.exe</SourceImage>
  <SourceImage condition="is">C:\Windows\system32\winlogon.exe</SourceImage>
  <SourceImage condition="is">C:\Windows\system32\audiodg.exe</SourceImage>
  <StartModule condition="is">C:\Windows\system32\kernel32.dll</StartModule>
</CreateRemoteThread>
```

```
<!--Event ID 10 ProcessAccess -->
<ProcessAccess onmatch="include">
</ProcessAccess>
```



```

Select Administrator: Windows PowerShell
-a----          5/29/2026    2:27 PM          5375 Sysmon_Configuration.xml

PS Y:\system_tools> .\Monitoring_with_Sysmon.ps1

Security warning
Run only scripts that you trust. While scripts from the internet can be useful, this script can potentially harm your computer. If you trust this script, use the Unblock
cmdlet to allow the script to run without this warning message. Do you want to run Y:\system_tools\Monitoring_with_Sysmon.ps1?
[D] Do not run [R] Run once [S] Suspend [?] Help (default is "D"): r
[05/29/2026 14:31:14 - EventID: 8]
CreateRemoteThread detected:
RuleName: -
UtcTime: 2026-05-29 07:31:14.084
SourceProcessGuid: {685cb1df-40c2-6a19-8b01-000000003200}
SourceProcessId: 6320
SourceImage: \\VBoxSrv\sharing_folder\system_tools\code_injection.exe
TargetProcessGuid: {685cb1df-3d2c-6a19-0801-000000003200}
TargetProcessId: 1140
TargetImage: C:\Windows\explorer.exe
NewThreadId: 5520
StartAddress: 0x0000000002490000
StartModule: -
StartFunction: -
SourceUser: PENTESTAD\Administrator
TargetUser: PENTESTAD\Administrator
-----

06/18/2025 09:04 PM 4,124,090 PROCMON.exe
06/18/2025 09:04 PM 8,480,560 Sysmon.exe
06/18/2025 09:04 PM 4,563,248 Sysmon64.exe
05/29/2026 02:27 PM 5,375 Sysmon_Configuration.xml
7 File(s) 17,460,709 bytes
2 Dir(s) 245,498,986,496 bytes free

Y:\system_tools>code_injection.exe
Explorer's PID = 1140

Y:\system_tools>
```

Case 2 - Detecting Credential Dumping

Detection rules

```
<Sysmon schemaversion="4.90">
  <EventFiltering>

    <ProcessAccess onmatch="include">
      <TargetImage condition="is">C:\Windows\System32\lsass.exe</TargetImage>
    </ProcessAccess>

  </EventFiltering>
</Sysmon>
```

```
<ProcessCreate onmatch="include">
  <Image condition="contains">mimikatz</Image>
</ProcessCreate>
```

```
<CreateRemoteThread onmatch="include">
</CreateRemoteThread>
```

ne

Type

new events available

Source

M

Sysmon

M

Sysmon

M

Sysmon

M

Sysmon

M

Sysmon

M

Sysmon

M

Sysmon

M

Sysmon

M

Sysmon

Sysmon/Operational

Logged: 5/29/2026 3

Task Category: Image load

"

s^C

Y:\mimikatz_trunk\x64>mimikatz.exe

.#####. mimikatz 2.2.0 (x64) #19041 Sep 19 2022 17:44:08

.## ^ ##. "A La Vie, A L'Amour" - (oe.eo)

/ \ ## /*** Benjamin DELPY `gentilkiwi' (benjamin@gentilkiwi.com)

\ / ## > https://blog.gentilkiwi.com/mimikatz

'## v ##' Vincent LE TOUX (vincent.letoux@gmail.com)

'#####' > https://pingcastle.com / https://mysmartlogon.com ***/

mimikatz # privilege::debug

Privilege '20' OK

mimikatz # sekurlsa::logonpasswords

Authentication Id : 0 ; 3150628 (00000000:00301324)

Session : Interactive from 1

User Name : DWM-1

Domain : Window Manager

Logon Server : (null)

Logon Time : 5/29/2026 3:02:46 PM

SID : S-1-5-90-0-1

msv :

[00000003] Primary

* Username : WINSERVER2022\$

* Domain : PENTESTAD

* NTLM : f95a3ed8d9dba01266000b476db9b5dc

* SHA1 : 902753b55722cdb90f3f6c071c1a07ae4987a118

tspkg :

wdigest :

* Username : WINSERVER2022\$

* Domain : PENTESTAD

* Password : (null)

kerberos :

* Username : WINSERVER2022\$

* Domain : PentestAD.local

* Password : 63 50 60 10 42 6c 5d ae 5d 08 39 b5 d7 80 9f a9 e0 f2 41 e1 2f

ee ff dd 65 2d ff cb c1 ac 6a bb 40 23 1c da 1f 63 eb be 49 db ee e7 94 37 3d 31 90

0e 7b 1d 30 49 9c 43 16 c7 b2 a1 40 2a 78 5d d3 f8 73 8c b6 c3 33 25 9b 0b fa dc 8-

4b aa c2 5f d3 5c fd cd bf 33 6d 28 ff d5 c5 b2 90 ec dc 4a 4a 34 be 53 7c 27 a5 20

86 db 11 13 f6 43 07 87 a4 00 1f 9b 0b 39 e8 ba cb e2 e8 c2 fc 22 96 af fa d3 d1 34

bb 98 61 72 19 4c a4 70 85 2b 87 05 91 ee 68 47 99 f3 20 84 5a ea 43 08 05 45 1f 40

ad 80 bb 2b 63 9d 52

ssp :

credman :

cloudap :

Filtered: Log: Microsoft-Windows-Sysmon/Operational; Source: ; Event ID: 8,10. Number of events: 113

Level	Date and Time	Source	Event ID	Task Category
Information	5/29/2026 3:19:33 PM	Sysmon	8	CreateRemot...
Information	5/29/2026 3:19:33 PM	Sysmon	8	CreateRemot...
Information	5/29/2026 3:18:59 PM	Sysmon	8	CreateRemot...
Information	5/29/2026 3:18:58 PM	Sysmon	8	CreateRemot...
Information	5/29/2026 3:18:28 PM	Sysmon	8	CreateRemot...
Information	5/29/2026 3:18:26 PM	Sysmon	8	CreateRemot...
Information	5/29/2026 3:12:12 PM	Sysmon	8	CreateRemot...
Information	5/29/2026 3:12:11 PM	Sysmon	8	CreateRemot...

Event 8, Sysmon

General Details

SourceProcessId: 472
SourceImage: C:\Windows\System32\csrss.exe
TargetProcessGuid: {685cb1df-4c06-6a19-0b01-00000003300}
TargetProcessId: 5328
TargetImage: \\VBoxSrv\sharing_folder\mimikatz_trunk\x64\mimikatz.exe

Log Name: Microsoft-Windows-Sysmon/Operational
Source: Sysmon Logged: 5/29/2026 3:19:33 PM
Event ID: 8 Task Category: CreateRemoteThread detected (rule: CreateRemoteThread)
Level: Information Keywords:
User: SYSTEM Computer: WinServer2022.PentestAD.local
OpCode: Info
More Information: [Event Log Online Help](#)

```
[05/29/2026 15:31:17 - EventID: 10]
Process accessed:
RuleName: -
UtcTime: 2026-05-29 08:31:17.497
SourceProcessGUID: {685cb1df-4ec8-6a19-1101-00000003300}
SourceProcessId: 5356
SourceThreadId: 3708
SourceImage: Y:\mimikatz_trunk\x64\mimikatz.exe
TargetProcessGUID: {685cb1df-4a98-6a19-0c00-00000003300}
TargetProcessId: 632
TargetImage: C:\Windows\system32\lsass.exe
GrantedAccess: 0x1010
CallTrace: C:\Windows\SYSTEM32\ntdll.dll+9f3b4|C:\Windows\System32\KERNELBASE.dll+2aafe|Y:\mimikatz_trunk\x64\mimikatz.exe+c291e|Y:\mimikatz_trunk\x64\mimikatz.exe+c2cf5|Y:\mimikatz_trunk\x64\mimikatz.exe+c285d|Y:\mimikatz_trunk\x64\mimikatz.exe+85a44|Y:\mimikatz_trunk\x64\mimikatz.exe+8587c|Y:\mimikatz_trunk\x64\mimikatz.exe+85647|Y:\mimikatz_trunk\x64\mimikatz.exe+c97a5|C:\Windows\System32\KERNEL32.DLL+14ed0|C:\Windows\SYSTEM32\ntdll.dll+7e39b
SourceUser: PENTESTAD\Administrator
TargetUser: NT AUTHORITY\SYSTEM
```

Case 3 - Detect Catching Active Directory Database Theft

Detection rules

```
<ProcessCreate onmatch="include">
  <Image condition="image">C:\Windows\System32\ntdsutil.exe</Image>
  <Image condition="image">C:\Windows\SysWOW64\ntdsutil.exe</Image>
</ProcessCreate>
```

```
<FileCreate onmatch="include">
  <TargetFilename condition="contains">NTDS.dit</TargetFilename>
  <TargetFilename condition="contains">ntds.dit</TargetFilename>
</FileCreate>
```

ntdsutil.exe 'ac i ntds' 'ifm' 'create full c:\temp' q q

```
Y:\mimikatz_trunk>powershell "ntdsutil.exe 'ac i ntds' 'ifm' 'create full y:\temp' q q"
```

```

Y:\mimikatz_trunk>powershell "ntdsutil.exe 'ac i ntds' 'ifm' 'create full y:\temp' q q"
C:\Windows\system32\ntdsutil.exe: ac i ntds
Active instance set to "ntds".
C:\Windows\system32\ntdsutil.exe: ifm
ifm: create full y:\temp
Creating snapshot...
Snapshot set {01ca260c-31da-43ab-b41b-18372ab16ac6} generated successfully.
Snapshot {da34b771-520d-4c41-b630-706eb98191af} mounted as C:\$SNAP_202605291538_VOLUMEC$\
Snapshot {da34b771-520d-4c41-b630-706eb98191af} is already mounted.
Initiating DEFRAGMENTATION mode...
Source Database: C:\$SNAP_202605291538_VOLUMEC$\Windows\NTDS\ntds.dit
Target Database: y:\temp\Active Directory\ntds.dit

Defragmentation Status ( omplete)

  0    10    20    30    40    50    60    70    80    90   100
|----|----|----|----|----|----|----|----|----|----|
|.....|
.....

Copying registry files...
Copying y:\temp\registry\SYSTEM
Copying y:\temp\registry\SECURITY
Snapshot {da34b771-520d-4c41-b630-706eb98191af} unmounted.
IFM media created successfully in y:\temp
ifm: q
C:\Windows\system32\ntdsutil.exe: q

```

Operational - Windows Event Viewer

Event Properties - Event 1, Sysmon

General Details

Process Create:
 RuleName: -
 UtcTime: 2026-05-29 08:59:47.459
 ProcessGuid: {685cb1df-5583-6a19-3401-000000003300}
 ProcessId: 2872
 Image: C:\Windows\System32\ntdsutil.exe
 FileVersion: 10.0.20348.469 (WinBuild.160101.0800)
 Description: NTSDS
 Product: Microsoft® Windows® Operating System
 Company: Microsoft Corporation
 OriginalFileName: ntdsutil.exe
 CommandLine: "C:\Windows\system32\ntdsutil.exe" "ac i ntds" ifm "create full y:\temp" q q
 CurrentDirectory: Y:\system tools\

Log Name: Microsoft-Windows-Sysmon/Operational

Source: Sysmon Logged: 5/29/2026 3:59:47 PM
 Event ID: 1 Task Category: Process Create (rule: ProcessCreate)
 Level: Information Keywords:
 User: SYSTEM Computer: WinServer2022.PentestAD.local
 OpCode: Info
 More Information: [Event Log Online Help](#)

Event Properties - Event 11, Sysmon

General Details

File created:
RuleName: -
UtcTime: 2026-05-29 09:07:07.928
ProcessGuid: {685cb1df-5735-6a19-4101-000000003300}
ProcessId: 2644
Image: C:\Windows\system32\ntdsutil.exe
TargetFilename: C:\temp\Active Directory\ntds.dit
CreationUtcTime: 2026-05-29 09:07:07.928
User: PENTESTAD\Administrator

Log Name: Microsoft-Windows-Sysmon/Operational
Source: Sysmon
Event ID: 11
Level: Information
User: SYSTEM
OpCode: Info
Logged: 5/29/2026 4:07:07 PM
Task Category: File created (rule: FileCreate)
Keywords:
Computer: WinServer2022.PentestAD.local
More Information: [Event Log Online Help](#)

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